

AI-Powered Requirement Analyzer

Objective

The aim of this project is to automate the initial stages of software development by analyzing high-level business requirements and converting them into:

- A structured list of system modules
- A proposed database schema in JSON-like format
- Pseudocode for core application features

This tool assists software engineers, analysts, and product teams in quickly translating abstract requirements into actionable design components.

Tools and Technologies

- **Programming Language:** Python
- **AI Model:** google/flan-t5-large from Hugging Face
- **Key Libraries:**
 - transformers (for text-to-text generation tasks)
 - torch (for running the model on CPU or GPU)

Functional Overview

1. Model Initialization

The application loads the Flan-T5 model and tokenizer once at runtime, ensuring efficient performance during repeated use.

2. Response Generation

A central function is used to pass prompt-based queries to the model, generating consistent and structured responses based on high-level requirements.

3. Requirement Analysis

The application prompts the model with three focused questions derived from the user input to produce:

- A breakdown of core software modules
- A database schema suitable for implementation
- Pseudocode that outlines major feature logic

4. CLI Interface

The program can be run directly from the terminal, where users input a requirement and receive organized analytical output. This design supports integration into developer workflows and automation pipelines.

Benefits

- Speeds up the software planning and documentation process
- Enhances accuracy and consistency in early-stage analysis
- Assists both technical and non-technical stakeholders in visualizing software structure
- Reduces dependency on manual documentation

Future Enhancements

- Integration with a web-based interface for real-time usage
- Support for exporting structured outputs into formats like PDF, Markdown, or DOCX
- Enhanced control over prompt templates to suit different industries
- Extension to include test cases, user stories, and API documentation generation

